# SAHAYAK - PRESENTATION GUIDE

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### LIVE DEMONSTRATION, TESTING AND CONCLUSION

Suggested speaking time: 4-5 minutes

Speaking script, slide cues, handover, troubleshooting and judge Q&A

## YOUR ROLE

You are the final speaker. Demonstrate the live website if it is working, explain what counts as a fair test, describe limitations and future plans, then give the conclusion and invite questions. Be ready with a backup if the camera or internet fails.

## SLIDE 12 - INTRODUCE THE DEMONSTRATION

Say: Thank you. I will show how someone can try SAHAYAK and explain how we should judge its output. This is a prototype, so the demonstration helps us see both what it can do and where it may fail.

Before opening the camera, make sure the audience can see the screen. Ask permission before using another person’s image or signing in a recorded demonstration.

## LIVE DEMO - STEP BY STEP

1. Open the SAHAYAK website and choose Live Studio.

2. Check internet access and confirm the backend is reachable. Allow camera permission when asked.

3. Place the hand clearly inside the camera frame and wait for the landmark overlay.

4. Perform a sign intended to be supported by the current model. Keep the hand visible and avoid rushing.

5. Point out the predicted label. Explain that a prediction is the model’s output, not a guarantee of correctness.

6. Try a second supported sign. If the model makes a mistake, acknowledge it and adjust the conditions once. Do not hide failed attempts.

## WHAT TO SAY WHILE THE DEMO RUNS

Say: The points drawn over the hand are landmarks detected by the browser. Those points are converted into numerical features. The backend model receives the features and returns a predicted label, which the website displays.

A good demonstration shows the process, but one or two successful predictions cannot establish overall accuracy. To know how reliable it is, we need repeatable testing on a larger set of examples.

## BACKUP PLAN IF THE DEMO FAILS

If the camera does not start: check browser permission and whether another app is using it. If it still fails, use a screenshot or architecture slide and explain the intended workflow.

If the backend does not respond: state that the prediction service is unavailable. Do not

say the model predicted a sign when no result returned.

If the overlay appears but a prediction is wrong: explain that landmark detection and classification are separate stages. Good landmarks do not guarantee a correct classification.

If internet is unavailable: explain that online backend inference and some translation or speech services need connectivity. Do not claim the complete application works offline.

## SLIDE 13 - HOW SHOULD WE TEST IT?

Say: We should collect repeated trials for each sign and record correct and incorrect predictions. Testing should include people and recording sessions not used during model training. We should vary lighting, distance, camera angle, signing speed, and hand visibility.

Per-sign results and a confusion matrix help show which signs are difficult to distinguish. We should measure response time separately. A fast prediction can still be wrong, so speed and correctness are different measures.

## SLIDE 14 - LIMITATIONS AND RESPONSIBLE USE

Say: The current model recognises only a limited vocabulary. It does not understand all ISL sentences, facial grammar, or regional variation. Landmark detection may fail when hands are hidden, out of frame, or poorly lit. The website is not a guaranteed communication service or a replacement for a qualified interpreter.

Responsible data collection matters. Samples should be collected with informed consent, stored carefully, and tested without leaking test examples into training. Users should understand what is processed or transmitted.

## SLIDE 15 - FUTURE IMPROVEMENTS

Say: Future work includes collecting a larger and more diverse dataset with consent, adding signs only after training and evaluation, improving sequence recognition, reporting results for each class, and testing usability with people who use Indian Sign Language. We also need to improve error handling and clearly communicate which features need internet access.

## FINAL CONCLUSION

Say: To conclude, SAHAYAK demonstrates a pipeline combining camera input, hand-landmark detection, machine learning, and a web interface. It is a starting point, not a finished universal translator. Useful assistive technology needs more than a working demo: it requires representative data, systematic testing, user feedback, privacy awareness, and honest reporting. Thank you for listening. We are happy to answer your questions.

Stage cue: Pause, face the judges, and invite questions.

## JUDGE QUESTIONS AND ANSWERS

Q: What happens when the model is wrong? A: We record the failure and investigate the input conditions and model behaviour instead of hiding it.

Q: Does the complete website work offline? A: No. Online backend inference and some translation or speech functions need connectivity. Some local phrase tools may work after loading, depending on the feature.

Q: How can the model improve? A: Collect diverse labelled examples, separate training and test people or sessions, evaluate per class, inspect errors, and retrain only after verification.

Q: How do you protect privacy? A: Request camera permission, explain what data is sent or stored, minimise retention, and obtain consent before collecting training samples.

Q: What is the current accuracy? A: Give a number only if there is a documented evaluation result. Otherwise, say it has not yet been reliably measured.

## FINAL DEMO CHECKLIST

Before judges arrive, test the camera, network connection, backend status, supported labels, and screen visibility. Keep a backup slide ready. Never invent accuracy statistics. If you do not know an answer, say: We have not measured that yet, and it is one of our planned tests.